A Comprehensive Study of the Chinese Warrants Bubble

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Abstract

We conduct the most comprehensive study to date of the Chinese warrants market in terms of the length of the sample period and the variety of the warrants investigated. During our sample period from August 2005 to September 2009, the underlying Chinese stock market peaked in October 2007, then crashed dramatically. Starting by the end of October 2008, the market recovered steadily. We examine the pricing of put warrants, covered call warrants and (uncovered) equity call warrants during such market swings. Our analysis indicates that Chinese warrants exhibited large pricing bubbles as their market prices significantly deviated from those implied by option pricing models. We show empirically that the bubble size is positively related to turnover, daily price change, and negatively related to the total number of warrants outstanding. These results are consistent with the prediction of the Greater Fool Theory: an investor buys securities without regard to their fundamental value, but with the hope of quickly selling them off at a higher price to another investor who might also be hoping to flip them quickly. Moreover, we find evidence that the stock price declines associated with the conversion of previously state-owned non-tradable shares into tradable ones dampened the bubble behavior for those companies' covered call warrants, relative to warrants on companies that did not participate in share conversions.

1 Introduction

Between August 2005 and September 2009, 18 put warrants and 37 call warrants were issued to the public in China. Among them, 39 were traded on the Shanghai Stock Exchange and 16 on the Shenzhen Stock Exchange. Despite being a young market, the Chinese warrants market has become one of the largest markets in the world in terms of trading volume. In the past few years, the warrant market has grown to account for over half of the total trading volume in China's financial markets. The average daily turnover rate was 207.42%. While the market is still developing, it is unique relative to other markets in several dimensions, among which the following two aspects are particularly relevant to the creation and extended existence of pricing bubbles in the Chinese warrants market.

(1) Legal Restriction on Short-Selling.

Both of China's warrant market and stock market do not allow short-selling in general. Only a few qualified investors can short sell stocks. Such restrictions prevent arbitrage even if the warrant or stock is overvalued. The market price discovery mechanism does not function well.

(2) Differential Regulations in China's Stock and Warrant Market.

First, China Securities Regulatory Commission imposes 10% daily price limit on both the upside and the downside from last trading day's closing for individual stocks. In the warrant market, price limit is much wider and rarely triggered. Therefore, speculators swarm into the warrant market.

Second, stock investors have to keep their stocks purchased today until the next trading day before selling, which is called the “T+1” rule; while warrant investors are able to sell on the same day the warrants they bought earlier during the day, which is called the “T+0” rule. A higher trading frequency is attributed to the “T+0” rule.

Third, transaction costs are lower for warrants than for stocks. The investors are exempt from paying stamp tax and registration fee when trading warrants. On the other hand, stamp tax is incurred for both buying and selling stocks. It changed over time, ranging from 0.1% to 0.3%. There is also a 0.1% registration fee on the proceeds when trading stocks. In addition, brokers usually charge a lower commission on warrant trading than on stock trading because of lager trading volume in the warrant market. Considering the inherent leverage provided by warrants, it is much cheaper to trade warrants than trading stocks.

The Chinese stock market started climbing higher since the beginning of 2006. However, with the large amount of new supply to the stock market due to the conversion of state-owned shares and the subprime crisis in the U.S., the China stock market crashed after reaching its peak by October 2007. For example, the Shanghai Composite Index tanked dramatically from the peak of 6,124 to the bottom of 1706.7. After the Chinese Government adopted stimulus measures to boost the national economy, the stock market started to rise steadily by the end of October 2008. Such dramatic swings in the market affected the prices of warrants significantly. Therefore, it is very important in our warrants study to use a long sample period from August 2005 to September 2009 that captures these market movements.

Xiong and Yu (2010) investigate the mispricing of Chinese put warrants. Because the underlying shares of these put warrants experienced the conversion of non-tradable government owned shares, share holders might value these put warrants greatly as a hedge due to the trading restrictions of the underlying shares. To refute such claims, they claim that the low correlation of the put warrants with the underlying stocks during the sample periods make the put warrants unlikely hedging instruments for the underlying stocks. However, this is not surprising given that the put warrants were out-of-the money during most of their sample. Had these puts became in-the-money, the correlation would have been much significant and the put warrants would have been good hedging instruments. In theory, put warrants and call warrants should behave differently under various market conditions. To get a complete picture of the warrants pricing behavior, we examine put warrants, covered call warrants, as well as (uncovered) equity call warrants.

To understand the determinants of bubble size in warrant prices, we use panel regressions to relate bubble size (the difference between market price and fundamental price implied by option pricing models) to variables suggested by the literature on financial bubbles. In particular we find that warrant bubble size is positively related to turnover, daily price change, and negatively related to the total number of warrants outstanding. These findings are consistent with the predictions of the Greater Fool Theory.

Greater Fool Theory states that an investor is willing to buy assets at a higher price than fundamental values, with the hope of quickly selling them off to another investor (the greater fool), who might also be hoping to flip them quickly at even higher prices. Camerer (1997) documents that John Maynard Keynes first proposed the Greater Fool Theory in the 1930s. Keynes believed that stock market is like a beauty contest. He took a beauty contest, very popular in England at the time, as an example. A newspaper would print 100 photographs, and people would say which six faces they liked the most. Everyone who picked the most popular face was automatically entered in a raffle, where they could win a prize. Keynes wrote, “It is not a case of choosing those faces which, to the best of one’s judgment, are really the prettiest, nor even those which average opinion genuinely thinks the prettiest. We have reached the third degree where we devote our intelligences to anticipating what average opinion expects the average opinion to be. And there are some, I believe, who practice the fourth, fifth and higher degrees.”

On one hand, warrant is new to many investors in the Chinese warrants market. The degree of understanding about warrants is quite different between sophisticated investors such as financial institutions and beginners. For instance, some investors chose to exercise when the strike prices are lower than the underlying stock prices at the maturity in quite a few put warrants. On the other hand, due to legal restrictions, investors cannot short-sell warrants even if they think warrant prices will eventually go down based on the underlying stock prices. Short-sell constraint prevents arbitrage and allows mispricing to persist for an extended period. Some investors long warrants at high prices even though some warrants are deep out of the money. Speculative motives and short-sell constraint together led to bubbles in warrant prices. Our study provides useful insights for market regulators to prevent future speculative bubbles in asset markets. Moreover, we find evidence that the stock price declines associated with the conversion of previously state-owned non-tradable shares into tradable ones dampened the bubble behavior for those companies' covered call warrants, relative to warrants on companies that did not participate in share conversions.

The rest of the paper is organized as follows. Section 2 describes the data used in our analysis. Section 3 uses one put warrant and one call warrant to illustrate the bubble phenomena. Section 4 outlines the empirical methodology. Section 5 discusses empirical results. Finally, we conclude in Section 6.

2. Data Descriptions

Chinese warrants issued between August 2005 and September 2009 included European style, American style, as well as Bermudan style warrants. We only include European warrants in our sample because their theoretical prices can be easily calculated using the Black-Scholes Model without worrying about the early exercise premium. As a result, we analyze a total of 39 listed warrants, among which 27 are call warrants and 12 are put warrants. Table 1, 2 and 3 provide some basic information for put warrants, covered call warrants, and (uncovered) equity warrants respectively. These warrants are adjusted for stock splits during their lifetime.

3 Examples of Pricing Bubbles

3.1 Kangmei Call Warrant

Kangmei call warrant was issued in May 2008 and expired in May 2009. It had an average daily turnover rate of 204.6 % and the maximum was 914.3 %. Its average daily trading volume was 1,109 million Yuan and the maximum was 5,304 million Yuan. Figure 1 shows the movements of the warrant price and the underlying stock price. Figure 2 illustrates the warrant price and its fundamental value. The area between the warrant price and its fundamental value represents the bubble size. During the middle and towards the end of its life, its fundamental value was close to zero, thus the warrant price consisted mostly of a pricing bubble. It is evident that the pricing bubble is significant and existed for an extended period of time. Figure 3 and 4 illustrate the daily turnover rates and the daily price changes during Kangmei's life time. Even on the last trading day, trading was still frequent and volatility was huge.

3.2 Nanhang Put Warrant

Nanhang put warrant was issued in July 2007 and expired in July 2008. It had an average daily turnover rate of 153 % and the maximum was 1,474.4 %. Its average daily trading volume was 10,185 million Yuan and the maximum was 45,419 million Yuan. Put warrant price should have a negative correlation with its underlying stock price. However, the correlation between the warrant price and its underlying stock price was 0.0707. The warrant price movements were slightly positively related to the underlying stock price movements. Figure 5 and 6 show the movements of the warrant price and its underlying stock price, and the movements of the warrant price and its fundamental value respectively. The area between the warrant price and the fundamental value measures the bubble size. 50 days after the issue date, the fundamental became close to zero, thus the warrant price consisted mostly of a pricing bubble. Figure7 and 8 illustrate the daily turnover rates and the daily price changes during Nanhang's life time. On the last trading day, the turnover rate was almost 1,500 %, indicating lots of speculative trading took place even when the price was close to zero.

4. **Methodology**

In this section, we empirically investigate the determinants of the pricing bubble in Chinese warrants.

The difference between a warrant's market price and its theoretic value proxies for the size of the pricing bubble. It is then used as the dependent variable in regressions. Since our sample is restricted to European style warrants, we can use Black-Scholes option pricing model to calculate the theoretic prices for the warrants. We have also tried other more sophisticated warrant pricing models but they do not alter our regression results in a significant way. The volatility input we use is one-year historical volatility. A potential issue of using the Black Scholes formula for valuing (uncovered) equity call warrants is the dilution effect caused by potential warrants exercise. However, according to Veld (2000), it should only become a concern if both the dilution factor is large AND the warrant is deep out of the money. This is not the case for the equity warrants in our sample. Therefore the approximation of using the Black Scholes formula is accurate enough for uncovered call warrants in this study.

As a warrant nears its maturity, its price converges to its final payoff. The time to maturity therefore has an impact on the size of the pricing bubble. So we include maturity fixed effects in our panel data regressions.

Consistent with the Greater Fool Theory, large pricing bubbles are more likely to exist if the asset is traded more frequently and its price exhibit larger movements[2], as the opportunity to sell the asset to the next investor at a higher price becomes more likely. In addition, the larger the size of the market, measure by the total number of tradable shares outstanding, the harder it is to generate a large pricing bubble, everything else being equal, as documented by Ofek and Richardson (2003) about the Internet bubble burst. Therefore we use daily turnover, daily price change (the absolute value of warrant closing price minus its opening price), size (measure by the total number of warrants outstanding) as three explanatory variables to explain warrants mispricing. In particular, we run the following regressions:

$$\text{Mispricing}_{it} = g_1 * \text{Turnover}_{it} + \text{Maturity Fixed Effect} + e_{it}. \quad (1)$$

$$\text{Mispricing}_{it} = k_1 * \text{Daily Price Change}_{it} + \text{Maturity Fixed Effect} + e_{it}. \quad (2)$$

$$\text{Mispricing}_{it} = m_1 * \text{Size}_{it} + \text{Maturity Fixed Effect} + e_{it}. \quad (3)$$

$$\text{Mispricing}_{it} = l_1 * \text{Turnover}_{it} + l_2 * \text{Daily Price Change}_{it} + \text{Maturity Fixed Effect} + e_{it}. \quad (4)$$

$$\text{Mispricing}_{it} = n_1 * \text{Turnover}_{it} + n_2 * \text{Daily Price Change}_{it} + n_3 * \text{Size}_{it} + \text{Maturity Fixed Effect} + e_{it}. \quad (5)$$

Mispricing_{it} is the difference between the closing price of warrant i on day t and its theoretic value calculated based on Black-Scholes option pricing model on day t,

[2] See Cochrane (2003)), see Lamont and Thaler (2003), Ofek and Richardson (2003) on the Internet bubble.

which is indexed by the number of trading days remaining, and e_{it} is the residual. Note that there is no intercept term in these regressions as it is absorbed by the Maturity Fixed Effect.

To investigate if the pricing bubbles are different for covered call warrants and uncovered call warrants, we include in our regressions a Covered Warrant Dummy. (It equals 1 if the warrant is a covered warrant, 0 otherwise). We run the following regressions for all call warrants in our sample:

$$Mispricing_{it} = p_1 * Turnover_{it} + p_2 * Daily\ Price\ Change_{it} + p_3 * Size_{it} + p_4 * Covered\ Warrant\ Dummy_{it} + Maturity\ Fixed\ Effect + e_{it} \quad (6)$$

5. Empirical Results

5.1 Put Warrants

Table 4 shows, when mispricing of the put warrants is regressed against turnover and daily price change individually, each produces positive and significant coefficients. The coefficients of the regression on turnover rate and daily price change together are also positive and significant. With size added to the previous regression, its coefficient is negative and significant. (In the single regression, mispricing is negatively related to size, though not statistically significant.)

5.2 Call Warrants

5.2.1 Call Warrants without Covered Warrants Dummy

Table 5 shows, for call warrants in our sample, when mispricing is regressed against turnover rate, daily price change individually, they each produce positive and significant coefficients. The coefficients in the multiple regression on turnover rate and daily price change are both positive and significant. As in the previous table for put warrants, with size itself or added to the above regressions, its coefficient is negative and significant.

5.2.2 Call Warrants with Covered Warrant Dummy

In Table 6 we include covered warrant dummy as an additional explanatory variable in the regressions. The coefficients for other variables remain consistent with those in the previous table. The coefficient for the covered warrant dummy is negative and significant in all the multiple regressions. This indicates that covered call warrants on average exhibit smaller pricing bubbles than uncovered equity call warrants when considered in isolation from the influence of other factors. Such result might have to do with the fact that all the covered call warrants in the sample were issued to tradable share owners by state-owned enterprises (who converted non-tradable shares into tradable) as compensation for potential share price decline due to a supply increase with the conversion. The price decline for such stocks resulted in less speculative interest in its warrants and therefore their call warrants price exhibited less bubble-like behavior.

6. Conclusion

We conduct the most comprehensive study to date of the Chinese warrants market in terms of the length of the sample period and the variety of the warrants investigated. In particular, we examine the pricing of put warrants, covered call warrants as well as (uncovered) equity call warrants from August 2005 to September 2009.

Our analysis indicates that the Chinese warrants market exhibited large pricing bubbles as their market prices significantly deviated from those implied by option pricing models. We show empirically that the bubble size is positively related to turnover, daily price change, and negatively related to the total number of warrants outstanding. These results are consistent with the prediction of the Greater Fool Theory: an investor buys securities without regard to their fundamental value, but with the hope of quickly selling them off at a higher price to another investor who might also be hoping to flip them quickly. Moreover, we find evidence that the stock price declines associated with the conversion of previously state-owned non-tradable shares into tradable ones dampened the bubble behavior for those companies' covered call warrants, relative to warrants on companies that did not participate in share conversions.

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Appendix

Table 1 Summary Information for 12 Put Warrants

This table shows, for each put warrant, its ticker, name, total number of warrants outstanding at the start of warrant trading, and trading period. This table also shows the closing price of the underlying stock on the first trading day, along with the initial warrant strike price and exercise ratio (number of shares of the underlying stock controlled per warrant).

Ticker	Name	Warrant Amount (Million shares)	Issue Information			Trading Period	
			Stock Price	Strike Price	Exercise Ratio	Begin	End
580999	WuGangJTP1	474	2.77	3.13	1	11/23/2005	11/15/2006
580994	YuanShuiCTP1	280	4.27	5	1	4/19/2006	2/5/2007
580996	HuChuangJTP1	568	11.85	13.6	1	3/7/2006	2/27/2007
580995	BaoGangJTP1	715	2.1	2.45	1	3/31/2006	3/23/2007
580993	WanHuaHXP1	85	16.42	13	1	4/27/2006	4/19/2007
38001	GangFanPGP1	233	3.3	4.85	1	12/5/2005	4/24/2007
580991	HaierJTP1	607	4.74	4.39	1	5/22/2006	5/9/2007
580992	YaGEQCP1	635	6.8	4.25	1	5/22/2006	5/14/2007
580990	MaotaiJCP1	432	48.39	30.3	0.25	5/30/2006	5/22/2007
580997	ZhaoHangCMP1	2241	6.37	5.65	1	3/2/2006	8/24/2007
38003	HuaLingJTP1	633	3.64	4.9	1	3/2/2006	2/22/2008
580989	NanhangJTP1	1400	8.99	7.43	0.5	6/21/2007	6/13/2008

Table 2 Summary Information for 9 Covered Call Warrants

This table shows, for each covered call warrant, its ticker, name, total number of warrants outstanding at the start of warrant trading, and trading period. This table also shows the closing price of the underlying stock on the first trading day, along with the initial warrant strike price and exercise ratio (number of shares of the underlying stock controlled per warrant).

Ticker	Name	Warrant Amount	Issue Information			Trading Period	
		(Million shares)	Stock Price	Strike Price	Exercise Ratio	Begin	End
580000	baogangJTB1	388	4.62	4.5	1	8/18/2005	8/30/2006
580001	wugangJTB1	474	2.65	2.9	1	11/23/2005	11/22/2006
30001	angangJTC1	113	1.61	3.6	1	12/5/2005	12/5/2006
580002	baogangJTB1	715	2.07	2	1	3/31/2006	3/30/2007
580003	hangangJTB1	926	5.1	2.8	1	4/5/2006	4/4/2007
580004	shouchuangJTB1	60	7.33	4.55	1	4/24/2006	4/23/2007
580005	wanhuaHXB1	57	28.93	9	1	4/27/2006	4/26/2007
580006	yageQCB1	91	14.39	3.8	1	5/22/2006	5/21/2007
580008	guodianJTB1	151	5.47	4.8	1	9/5/2006	9/4/2007

Table 3 Summary Information for 18 (Uncovered) Equity Call Warrants

This table shows, for each covered call warrant, its ticker, name, total number of warrants outstanding at the start of warrant trading, and trading period. This table also shows the closing price of the underlying stock on the first trading day, along with the initial warrant strike price and exercise ratio (number of shares of the underlying stock controlled per warrant).

Ticker	Name	Warrant Amount	Issue Information			Trading Period	
		(Million shares)	Stock Price	Strike Price	Exercise Ratio	Begin	End
580007	changdianCWB1	1228	7.28	5.5	1	5/25/2006	5/24/2007
580009	yiliCWB1	155	18.2	8	1	11/15/2006	11/14/2007
580011	zhonghuaCWB1	180	7.22	6.58	1	12/18/2006	12/17/2007
580015	rizhaoCWB1	62	13.2	14.25	1	12/3/2007	12/2/2008
580017	ganyueCWB1	56	16.98	8.4	1	3/7/2008	3/6/2009
580012	yunhuaCWB1	54	21.9	18.23	1	3/8/2007	3/7/2009
580020	shanggangCWB1	292	2.76	10.2	1	4/17/2007	4/16/2009
580013	wugangCWB1	728	10.73	10.77	0.5	5/26/2008	5/25/2009
580023	kangmeiCWB1	167	9.68	5.5	1	7/18/2008	7/17/2009
580018	zhongyuanCWB1	51	41.01	35.5	0.5	9/25/2007	9/24/2009
580021	qingpiCWB1	105	22.9	13.85	1	10/30/2007	10/29/2009
580014	shengaoCWB1	108	12.32	27.43	1	1/8/2008	1/7/2010
580016	shangqiCWB1	227	27.5	9.19	0.5	7/11/2008	1/10/2010
580025	gezhouCWB1	302	8.08	78.13	0.5	2/22/2008	2/21/2010
580019	shihuaCWB1	3030	17.38	19.68	0.5	3/4/2008	3/3/2010
580022	guodianCWB1	427	7.36	7.5	1	5/22/2008	5/21/2010
580024	baogangCWB1	1600	8.43	12.5	0.5	7/4/2008	7/3/2010
580026	jiangtongCWB1	1761	12.59	15.44	0.25	10/10/2008	10/9/2010

Table 4 Determinants of the Mispricing of Put Warrants

This table reports the panel regression results of (1) Mispricing (daily difference between the market price and theoretical price) of put warrants on Turnover (daily warrant's turnover), Price change (absolute value of daily warrant closing price minus opening price), and Size (daily total number of warrants outstanding) in column 1, 2, 3 respectively; (2) Mispricing on Turnover and Price change together in column 4; (3) Mispricing on Turnover, Price change and Size together in column 5. All regressions include warrant maturity fixed effect.

	1	2	3	4	5
Turnover	0.1200			0.0680	0.0582
(t-stat)	38.0374			20.6516	18.6054
Price change		1.6074		1.4292	1.4595
(t-stat)		68.3516		57.0457	59.5891
Size			-0.0030		-0.0074
(t-stat)			-3.033		-8.1037

Table 5 Determinants of the Mispricing of Call warrants

This table reports the panel regression results of (1) Mispricing (daily difference between the market price and theoretical price) of call warrants on Turnover (daily warrants turnover), Price change (absolute value of daily warrant closing price minus opening price), and Size(daily total number of warrants outstanding) in column 1, 2, 3 respectively; (2) Mispricing on Turnover and Price change together in column 4; (3) Mispricing on Turnover, Price change and Size together in column 5. All regressions include warrant maturity fixed effect.

	1	2	3	4	5
Turnover	0.3800			0.3917	0.3213
(t-stat)	42.0072			43.8836	35.7864
Price change		0.6673		0.6925	0.8372
(t-stat)		37.5403		39.6161	47.5256
Size			-1.1325		-0.9698
(t-stat)			-83.0714		-39.6680

Table 6 Determinants of the Mispricing of Call Warrants with Covered Warrants Dummy

This table reports the panel regression results of (1) Mispricing (daily difference between the market price and theoretical price) on Turnover (daily warrants turnover), Price change (absolute value of daily warrant closing price minus opening price) and Covered warrants dummy (A dummy variable that equals 1 if it is a covered call and 0 otherwise) together in column 1; (2) Mispricing on Turnover, Price change, Size (daily total number of warrants outstanding) and Covered warrants dummy together in column 2. All regressions include warrant maturity fixed effect.

	1	2
Turnover	0.3213	0.3215
(t-stat)	35.7864	36.0978
Price change	0.8372	0.5509
(t-stat)	47.5256	31.9237
Size		-1.0409
(t-stat)		-76.3706
Covered warrant dummy	-0.9698	-0.8832
(t-stat)	-39.6680	-36.4001

Figure 1 Kangmei Call Warrant Prices and Its Underlying Stock Prices

This figure shows both the daily market prices of Kangmei call warrant and its daily underlying stock prices from 5/26/08 to 5/18/09.

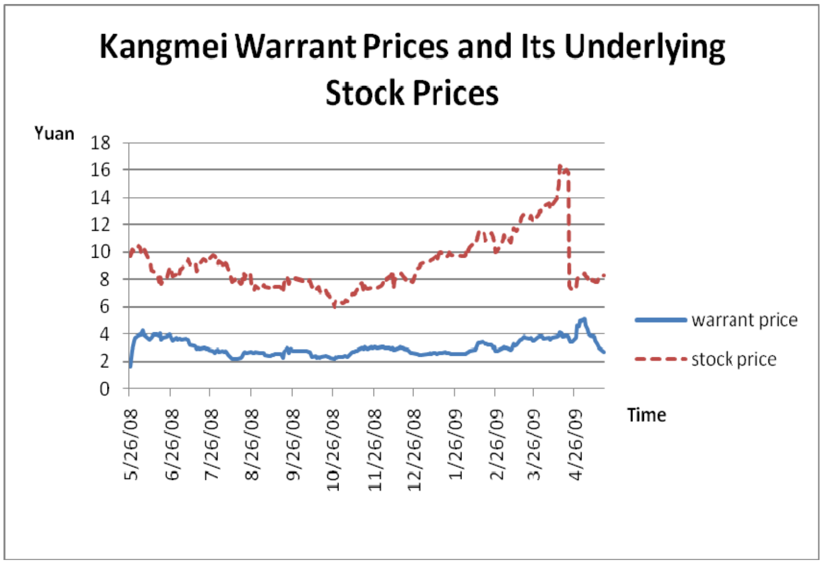

Figure 2 Kangmei Call Warrant and Its Fundamental Values

This figure show the daily Kangmei call warrant prices and it daily theoretic price from 5/26/08 to 5/18/09.

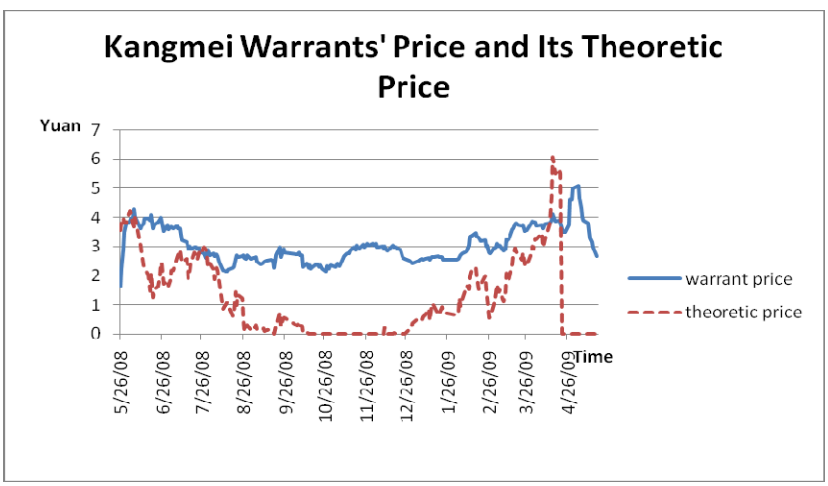

Figure 3 Kangmei Daily Turnover Rate

This figure shows the daily turnover rate of Kangmei call warrant from 5/26/08 to 5/18/09.

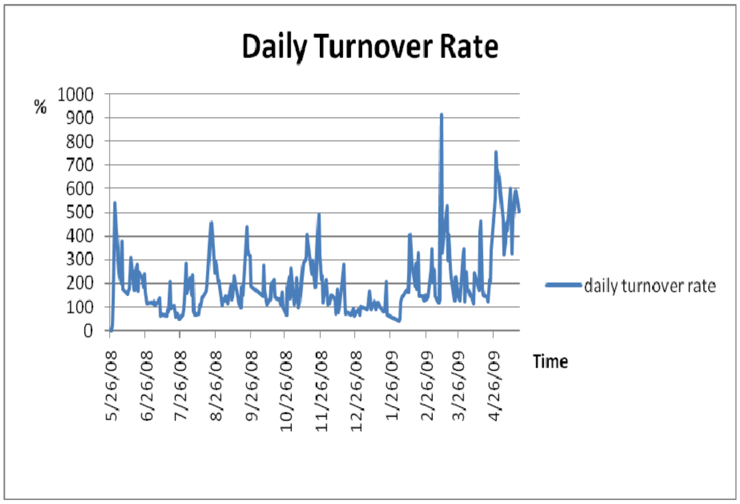

Figure 4 Kangmei Daily Price change

This figure shows the daily price change of Kangmei call warrant from 5/26/08 to 5/18/09.

Daily price change is defined as the absolute value of daily closing price minus opening price.

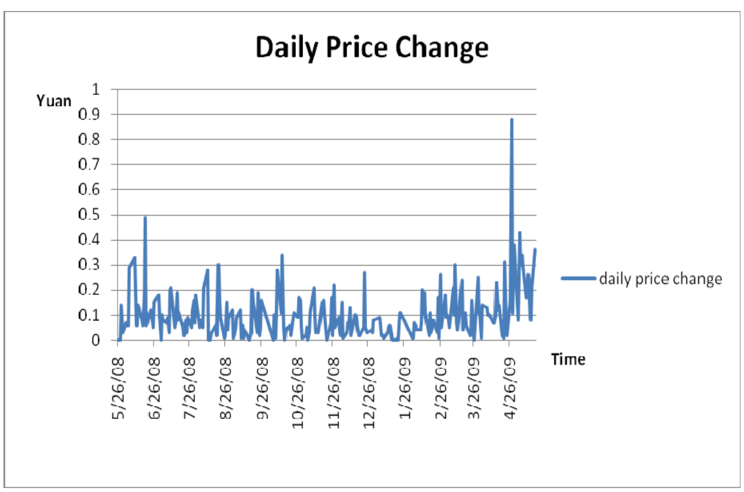

Figure 5 Nanhang Put Warrant Prices and Its Underlying Stock Prices

This figure shows both the daily market prices of Nanhang put warrant and its daily underlying stock prices from 6/21/07 to 6/11/08.

Figure 6 Nanhang Put Warrant and Its Theoretic Values

This figure show the daily Nanhang put warrant prices and it daily theoretic price from 6/21/07 to 6/11/08.

Figure 7 Nanhang Daily Turnover Rate

This figure shows the daily turnover rate of Nanhang put warrant from 6/21/07 to 6/11/08.

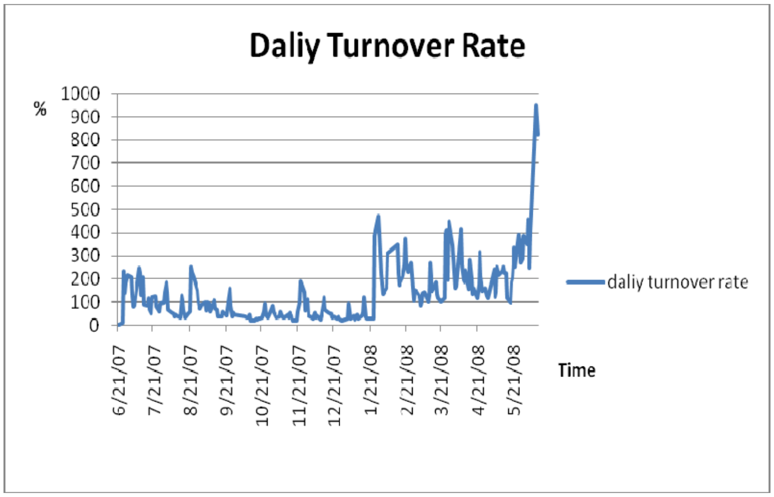

Figure 8 Nanhang Daily Price change

This figure shows the daily price change of Nanhang put warrant from 6/21/07 to 6/11/08.

Daily price change is defined as the absolute value of daily closing price minus opening price.